

Adaptive Personal Cooling System



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Project Overview

The Problem

- & Nearly 500,000 deaths annually from hyperthermia; core temps above 40 °C are a medical emergency
- & Bedridden, elderly, and ill individuals have diminished thermoregulatory responses (sweating, vasodilation)
- & Standard fixed-speed fans cannot respond dynamically to changing thermal conditions

What is the APCS?

- & A prototype embedded system that continuously measures ambient temperature and automatically adjusts fan speed via PID control
- & Built on the TM4C123GH6PM (ARM Cortex-M4F) microcontroller; interfaces DS18B20 sensor, potentiometer, fan, and I²C LCD

Main Objective

- & Provide automated, responsive personal cooling that adapts in real time — something a conventional fan cannot do

Design Requirements & Goals

Performance Targets

- & Regulate ambient temperature within $\pm 1-2$ °C of the user-selected setpoint (20–60 °C range)
- & Control loop update rate: ~2 seconds; DS18B20 12-bit resolution (0.0625 °C/step)
- & Fan response from setpoint change to observable airflow: ≤ 3 seconds

Constraints

- & Platform: TM4C123GH6PM LaunchPad; toolchain: Keil MDK-ARM with CMSIS device pack
- & Components: off-the-shelf only (DS18B20, 10 k Ω potentiometer, 4-wire 12 V PC fan, I²C LCD)
- & Firmware in 8 modular C files; no wireless, no battery, no multi-sensor architecture

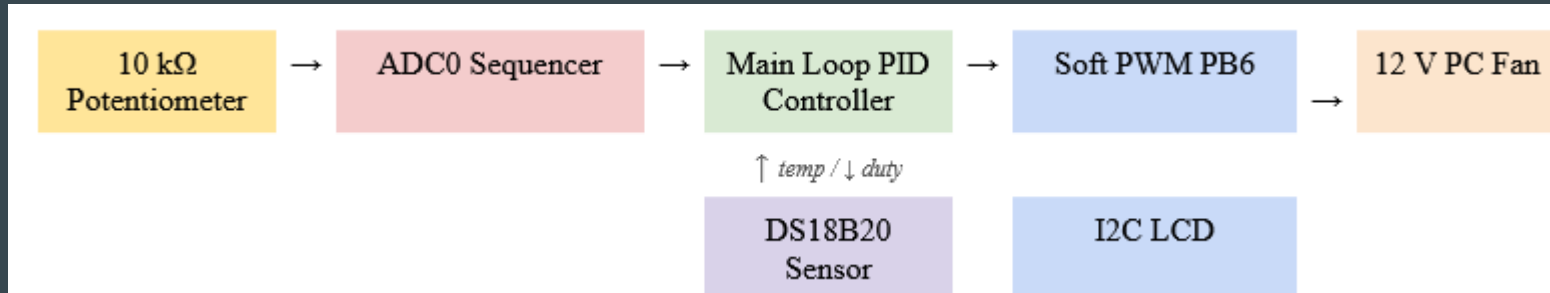
Success Criteria

- & PID controller converges to setpoint, fan operates without stall, LCD provides real-time feedback
- & System gracefully recovers from sensor loss without requiring a hard reset

Design Process & Architecture

System Block Diagram

- & Potentiometer → ADC0 Sequencer 3 → Main Loop PID Controller → Soft PWM PB6 → 12 V PC Fan
- & DS18B20 (1-Wire, PE1) feeds temperature into the PID; I²C LCD (PB2/PB3) shows live status



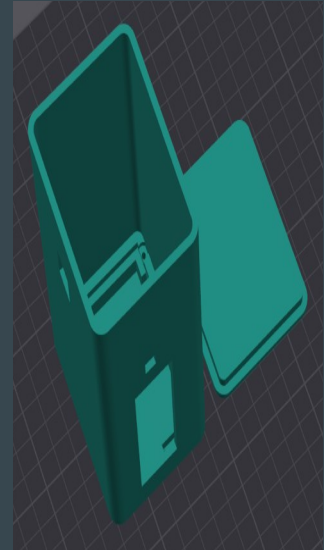
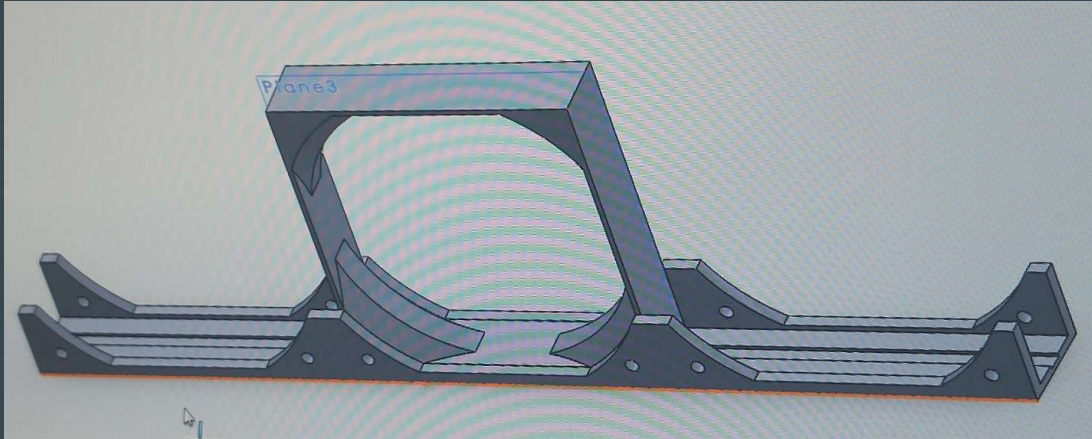
Firmware Architecture (8 modules)

- & main.c — control loop; adc.c — potentiometer; ds18b20.c — 1-Wire driver; pwm.c — fan drive
- & pid.c — discrete PID with anti-windup; lcd.c — I²C display; delay.c — SysTick timing; utils.c — string helpers

Design Process & Architecture

Hardware Design

- & SolidWorks rail mount bracket for the PC fan; 3D-printed enclosure with removable lid houses LaunchPad, LCD, and wiring
- & Potentiometer and LCD accessible through front-panel cutout



Build Process & Components

Key Components

- & TM4C123GH6PM LaunchPad — 80 MHz ARM Cortex-M4F, 256 KB flash, 32 KB SRAM
- & Dallas DS18B20 — ± 0.5 °C accuracy, 12-bit resolution, 1-Wire interface on PE1
- & 10 k Ω linear potentiometer — maps 0–3.3 V to 20–60 °C setpoint via ADC0
- & 4-wire 12 V PC fan — dedicated PWM input; hardware PWM on PB6 at 25 kHz
- & 16×2 I²C LCD (addr 0x28) — real-time display of temperature, setpoint, and duty cycle

Assembly Steps

- & Breadboard wiring: DS18B20 on PE1 with 5 k Ω pull-up; potentiometer wiper to PE3; fan PWM to PB6
- & SolidWorks CAD → 3D-print enclosure → mount LaunchPad, LCD, and fan rail bracket
- & Compile with Keil MDK-ARM; flash via on-board ICDI debugger; verify heartbeat LED on PF1

Challenges & Problem Solving

1. Push-Button vs. PID Design

- & Course spec required buttons; continuous potentiometer chosen instead — smoother PID setpoint, better for mobility-limited users

2. DS18B20 Power-On Default (85.0 °C)

- & Sensor pre-loads 0x0550 at startup; firmware detects sentinel, holds PID at zero until real conversion completes

3. 1-Wire Timing Sensitivity

- & Reset pulse $\geq 480 \mu\text{s}$; write-1 = $6 \mu\text{s}$, write-0 = $60 \mu\text{s}$; calibrated delayUs() against logic analyzer for reliable presence detection

4. Fan Stall Below 90% Duty

- & 10 Hz PWM on-time too short at low duty; MIN_DUTY clamped at 90%; 500 ms kick-start at 100% overcomes static inertia

5. ADC Noise on Potentiometer

- & ± 5 – 10 LSB jitter even with $64\times$ hardware oversampling; added 8-sample software average to reduce noise below ± 1 LSB

6. I²C Initialization Order

- & LCD corruption when I²C0 init ran before SysTick; fixed by calling SysTick_Init() first in main.c

Testing & Results

System Performance

- & PID converged to within $\pm 1\text{--}2$ °C of setpoint under steady-state lab conditions
- & Fan response latency: ~2 seconds (750 ms DS18B20 conversion + 20×100 ms PWM cycle)
- & Kick-start eliminated cold-start stall: 100% success vs. ~60% stall rate without it

Fault Tolerance

- & Sensor disconnect detected within one loop cycle; fan duty set to zero; re-acquisition automatic upon reconnection
- & Heartbeat LED toggled every loop pass — no watchdog violations or lockups observed in all test sessions

LCD & Display

- & Real-time display of current temperature, setpoint, and fan duty % confirmed accurate across full 20–60 °C range
- & Three status codes: normal operation, no-sensor fault, and startup/converting

Limitations

- & Minor oscillation due to thermal inertia of air mass and coarse 10 Hz PWM update rate

Live Demonstration

What to Watch For

- & LCD Line 1: current ambient temperature (from DS18B20) and fan duty % updating in real time
- & LCD Line 2: setpoint temperature — rotate potentiometer to change target
- & Fan speed visibly increases as temperature rises above setpoint; slows or stops when below

Demo Steps

- & 1. Power on — observe splash screen, then sensor detection and first reading on LCD
- & 2. Rotate potentiometer to lower setpoint below ambient — fan spins up, duty shown on display
- & 3. Rotate to raise setpoint above ambient — fan slows; observe kick-start on next spin-up
- & 4. Briefly disconnect DS18B20 wire — LCD shows NO SENSOR; reconnect and watch auto-recovery

Backup

- & Video recording of full demo available if live hardware cannot be presented

Future Improvements

Hardware Upgrades

- & Replace software PWM with TM4C123GH6PM hardware PWM module (25 kHz) for finer duty resolution and decoupled loop timing
- & Replace ambient DS18B20 with a skin-contact NTC thermistor patch for direct body temperature monitoring
- & Add Bluetooth Low Energy (BLE) module for smartphone setpoint control — critical for bedridden users

Software & Control

- & Implement adaptive PID gain scheduling: larger K_p at high error, smaller near setpoint to reduce oscillation
- & Add multi-sensor support for zone-based cooling in larger enclosures

Productization

- & Formal DS18B20 calibration against NIST-traceable reference (prerequisite for clinical deployment)
- & Battery-powered operation and form-factor miniaturization for wearable/portable applications

Conclusion

What Was Built

- & A fully functional closed-loop personal cooling system on the TM4C123GH6PM using DS18B20, PID, software PWM, and I²C LCD
- & Firmware organized across 8 modular C source files; compiled with Keil MDK-ARM

Was It Successful?

- & Yes — PID converged within $\pm 1\text{--}2$ °C; fan operated without stall; system recovered from sensor faults automatically
- & Validates PID-based closed-loop control on resource-constrained ARM Cortex-M4 hardware for thermal management

Key Takeaways

- & Hands-on experience with 1-Wire protocol, I²C initialization, discrete PID tuning, and motor drive behavior
- & Prototype establishes a foundation for a medically meaningful personal cooling device with further development

Questions?

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Thank You!!